
Quick Context on the Bubble

- **Google's "Willow" announcement (Dec 9)** suddenly made quantum computing the new "hot" sector.
- Similar pattern to other speculative manias (fusion, memecoins).
- My view: these stocks are humorously overvalued, akin to the dot-com bubble.

Core Reasons for Overvaluation

- **Investor Desperation for Quantum Exposure:**
  - Most "real" quantum players (e.g., Quantinuum, PsiQuantum) remain private.
  - Retail investors (and some institutions) jump on whatever public ticker has "quantum" in it, hoping not to miss out.
- **Specialized Machines, Not General-Purpose Computers:**
  - Quantum computers address very specific problems, not everyday computing tasks.
  - The hype ignores how narrow these use cases might be.

The Quantum Basics

- **Basic Idea:**
  - Proposed by Feynman and Deutsch in the 1980s.
  - Quantum computers exploit quantum phenomena (superposition, entanglement) to solve certain problems faster than classical computers.
  - Important to note: They are specialized machines, not general-purpose computers for running your favorite video games.
- **Key Algorithms:**
  - **Shor's Algorithm:** Fast factoring, crucial for cryptography.
  - **Grover's Algorithm:** A faster approach to 'search' problems, but harder to understand and implement.
- **The Real Challenge—Scale:**
  - Need enough logical qubits to outpace classical machines.
  - Currently, nobody is anywhere close to building large-scale, fault-tolerant quantum computers.
  - Biggest Firm on the list has a 121 physical Qubit processor.
  - According to specialists a few 1000 logical qubits are needed to be efficient against traditional computers —To get to one logical qubit one needs roughly 1000 physical qubits

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Stock Overviews & Why I'm Skeptical

1. **D-Wave Systems (QBTS)**

    - **Annealing vs. “Real” Quantum:**
        - They sell “quantum annealers,” which have limited applications and aren’t considered “universal” quantum computers.
    - **Bleak Financial History:**
        - Over 20 years with virtually no commercial success.
        - Revenue: $9M/year; Losses: $100M/year; Cash: $25M left as of Q3 2024.
    - **SPAC & Post-Google Spike:**
        - Went public via SPAC in 2022, hovered around $1 before Willow news.
        - Jumped to $9+ afterward (roughly $2.5B market cap).
    - **Valuation Is Ridiculous:** ~288× sales, no meaningful growth. Likely heading back to $1.

2. **Rigetti Computing (RGTI)**

    - **History & Burn Rate:**
        - Founded 2013 by Chad Rigetti (left recently).
        - Has spent $402 million since inception.
    - **Revenues Falling:**
        - 2022: $13M
        - 2023: $12M
        - 2024E: $12M (flat/declining).
    - **Desperate Funding Rounds:**
        - Did a SPAC, then sold stock at $1.53 (ATM) and $2.00 (registered direct).
        - After Willow mania, stock soared to $18.66.
    - **What They Do & Why It’s a Problem:**
        - They make “real” quantum computers using superconducting qubits (similar to Google/IBM).
        - Widely regarded as a “third-tier” competitor: their machines are too small and error-prone.
        - Ankaa-3 (84 qubits) is announced, but the predecessor (Ankaa-2) is offline on AWS Braket.
    - **Example Calculation (NSA Budget Scenario):**
        - NSA total budget is around $10–15B, maybe $2B for computing.
        - Suppose half goes to quantum ($1B). Then imagine Rigetti gets **50%** of that ($500M).
        - At a 50% net margin, discounted at 12–25% over 5 years, you get an NPV of **$1 to $4 per share**.
            - Current stock is **$18+**.
        - Another (unrealistic) scenario: $2B in revenue by **2035**, 50% margin. Discount yields **$3 to $14**/share at best.
        - Conclusion: even in best case scenario price is not justified at all. Margin of safety for shorts is therefore given

3. **IonQ (IONQ)**

    - **Trapped Ion Approach:**
        - Uses ions (instead of superconducting circuits) at room temperature but under ultra-low pressure + lasers.
        - Academically interesting, but industry skepticism remains (too finicky to scale).
    - **Appears Bigger, Still Behind:**
        - Has more revenue, mostly from government contracts (e.g., Air Force).

        - On AWS Braket, their best machine (Forte 1) is 36 qubits and often offline for maintenance.
    - **Still Behind Heavyweights:**
        - IBM, Google, Quantinuum, PsiQuantum are all further ahead in scale and reliability.
    - **Why the Stock Is High:**
        - Retail sees IonQ as one of the only "pure" quantum plays on the market.

4. **Quantum Computing Inc (QUBT)**

    - Essentially a worthless "scam" company riding the quantum name.
    - Market focuses on them solely because they have "quantum" in the name.

- **Other Stocks Investors Confuse for "Quantum":**
    - Quantum (QMCO), Quantum-Si (QSI), SealSQ (LAES).
    - They're not quantum computing players but have "quantum" in their name or marketing.

Broader Perspective

- **Jensen Huang (NVIDIA) Insight:**
    - Quantum computers might be **15–30 years** from being genuinely useful.
    - They'll only help with very specialized, combinatorial problems (not the main data/computing challenges).
- **Real Market Size for Quantum?**
    - Could be akin to pharma modeling software (e.g., Schrödinger, SDGR at $100M/year).
    - Or specialized lab equipment (e.g., Bruker, BRKR, a $9B company).
    - Even if quantum "works," it becomes a niche tool, not an all-purpose revolution.
- **Cryptography Angle:**
    - Shor's algorithm eventually could break RSA/ECDSA, but that still doesn't justify these sky-high valuations.

The Thesis

- **Why These Stocks Are So Overblown:**
    - **Investors are desperate for quantum exposure** yet the real innovators are still private.
    - Retail (and some institutional) folks pour money into any public ticker with "quantum" in the name, ignoring fundamentals.
- **Position:**
    - Short on D-Wave (QBTS), Rigetti (RGTI), IonQ (IONQ), and Quantum Computing Inc (QUBT).
    - no significant quantum impact for Alphabet (GOOG) or IBM (IBM) anytime soon.

Final Valuations (Discounted NPV)

Rigetti (RGTI): $18.37 – fair value: $1.00 IonQ (IONQ): $49.59 – fair value: $11.23 D-Wave (QBTS): $9.55 - fair value: $0.01 ** **Quantum Computing Inc (QUBT): $17.49 - fair value: $0.01**